

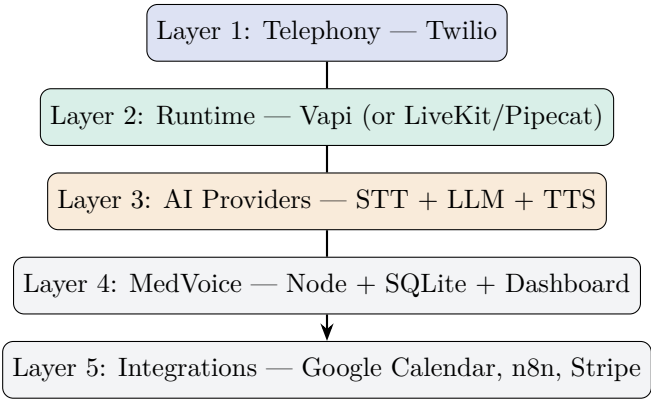
MedVoice AI

External Resources & Provider Stack

Infrastructure Planning

June 2026

1 Stack Architecture



2 Required External Services

2.1 1. Voice Runtime (choose one)

Provider	Role	Cost	HIPAA
Vapi (recommended MVP)	Call orchestration	\$0.05/min	BAA available
LiveKit Agents	Orchestration + infra	\$0.01/min cloud; self-host free	Scale tier \$500/mo
Pipecat Cloud	Pipeline framework	~\$0.01/min; OSS self-host	Self-host control
Retell AI	Alternative runtime	\$0.07–0.12/min	Enterprise

2.2 2. Speech-to-Text (STT)

Provider	Model	Cost/min	Notes
Deepgram (recommended)	Nova-2 Medical	~\$0.004–0.02	Best latency; medical vocab
AssemblyAI	Universal-2	~\$0.015	Strong accuracy
OpenAI Whisper API	whisper-1	~\$0.006	Simple; higher latency
Google Cloud STT	Medical dictation	Custom	HIPAA on GCP

2.3 3. Large Language Model (LLM)

Provider	Model	Cost	Use Case
OpenAI	GPT-4o mini	~\$0.003–0.01/min	Production phone (fast, cheap)
OpenAI	GPT-4o	Higher	Complex triage
Groq	Llama 3.1 70B	Very low	High-volume cost savings
Anthropic	Claude 3.5 Sonnet	Mid	Long context, safety
Ollama (current dev)	llama3.1:8b	Free local	Dev/demo only — not production

2.4 4. Text-to-Speech (TTS)

Provider	Voice Tier	Cost	Notes
ElevenLabs (recommended)	Flash v2.5	~\$0.01–0.07/min	Natural; clinic-branded voice
OpenAI TTS	tts-1	~\$0.015/min	Good quality; simple
Cartesia	Sonic	Low latency	Real-time optimized
PlayHT	Standard	~\$0.07/min	Premium voices
Azure Neural TTS	Standard	~\$0.02/min	Enterprise HIPAA path

2.5 5. Telephony

Provider	Cost	Notes
Twilio (recommended)	~\$1.15/mo per number + \$0.008–0.02/min	BYO into Vapi; SMS for confirmations
Vonage	Comparable	Alternative carrier
Telnyx	Lower per-min	Cost-sensitive scale

2.6 6. Calendar & Scheduling

Service	Status in MedVoice	Notes
Google Calendar API	Implemented (<code>calendar.js</code>)	Service account JWT; enable per tenant
Calendly	Config URL only	Fallback link in widget
athenahealth / Epic	Not started	Enterprise EHR — Phase 4+

2.7 7. Automation & CRM

Service	Role	Status
n8n	Webhook consumer for leads	Configured in <code>.env.example</code>
HubSpot / Salesforce	CRM sync	Not implemented
Zapier	No-code alternative	Not implemented

2.8 8. Payments & Billing

Service	Role	Status
Stripe	Subscriptions + usage billing	Not implemented
Stripe Tax	Sales tax automation	Not implemented

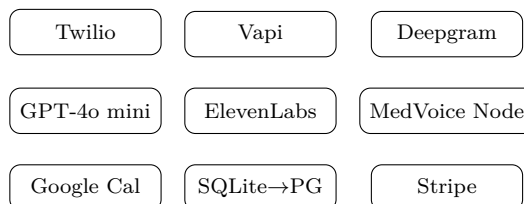
2.9 9. Hosting & Compliance

Service	Role	HIPAA
Railway / Render	Node backend hosting	Add BAA
AWS (HIPAA)	Production PHI workload	HIPAA-eligible services
Vercel / Netlify	Dashboard static hosting	No PHI on edge
Sentry	Error monitoring	Configure PHI scrubbing
Datadog	APM + logs	Enterprise BAA

3 Account Checklist

#	Account	Priority
1	Vapi (dashboard.vapi.ai)	P0 — phone MVP
2	Twilio	P0 — numbers + SMS
3	Deepgram	P0 — STT
4	OpenAI	P0 — LLM
5	ElevenLabs	P0 — TTS
6	Google Cloud (Calendar SA)	P0 — booking
7	Stripe	P1 — billing
8	Railway/AWS HIPAA	P1 — production
9	Sentry	P2 — monitoring
10	n8n (or self-host)	P2 — automation

4 Recommended Production Stack



All-in cost estimate: \$0.20–0.28/min (mid-tier stack) × monthly minutes + \$50–500 platform overhead.

5 BYOK Advantage

Bring-your-own-keys means: you hold vendor relationships, negotiate volume discounts, and sign BAAs directly—critical for medical sales. MedVoice margin sits in the subscription layer above raw provider cost.